Practice questions 3, Soft Eng 3MX3, Dr. M. v. Mohrenschildt

1 Short Questions

1. What is the minimal sampling frequency rounded to an integer that is needed such that we can sample and reconstruct the signal
$$x(t) = \cos(50\pi t) + \cos(100\pi t)$$

2 Difference Equations to State Space Equation

Given a system by its frequency response
$$H(\omega) = \frac{1}{1 - .5e^{-i\omega}}$$
Determine the $[A, B, C, D]$ representation of the system.

3 Z and Laplace Transform

1. Determine if the system
$$\ddot{y} + 4\dot{y} - 2y = x$$
is stable. (Compute the Laplace transform, compute the poles, and decide if stable).

2. The Z-transform is given by
$$\hat{X}(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$
Compute the Z-transform of the signal $x(n) = \delta(n) + \delta(n-1)$.

4 Filter Design

A digital system is running at the sampling frequency $f_s = 2'000Hz$. Construct a minimal order FIR filter (give the difference equation) that filters (removes) sinodials with the frequency $f = 500Hz$ and filters any constant signal. The filter should be normalized to have a gain of 1 at $f = 1'000Hz$.

5 Frequency response to Differential equation - Impulse response

After designing a filter we obtained the frequency response
$$H(\omega) = \frac{1}{1 + \frac{1}{2}e^{-i\omega}}$$
What is the corresponding difference equation and impulse response ?

6 Impulse Response

Given the impulse response
$$h(n) = \begin{cases} 1 & n = 0 \\ 2 & n = 1 \\ 3 & n = 2 \\ 0 & \text{otherwise} \end{cases}$$
Determine the difference equation of the system, then use convolution to compute the output to the input
$$x(n) = \delta(n) + \delta(n-1) + \delta(n-3)$$

7 Fourier Series

The Fourier Series of a continuous p-periodic signal is given my

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{i\omega_0 kt}$$

where

$$X_k = \frac{1}{p}\int_0^p x(t)e^{-i\omega_0 kt}dt$$

Compute X_0, X_1 for the signal

$$x(t) = \begin{cases} 1 & 0 \leq t < \pi \\ -1 & \pi \leq t < 2\pi \end{cases}$$

Make sure you have the right ω_0!

8 Brick-wall Filter

The CTFT if given by:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{i\omega t}d\omega \quad X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt$$

Given a continuous system with the frequency response

$$H(\omega) = \begin{cases} 1 & |\omega| \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Filter the signal $x(t) = \delta(t-2)$. (You first have to compute $X(\omega)$ then filter and then transform back!)

9 Impulse Response to Frequency Response

(10 points) Use the DTFT or the CTFT to determine the frequency response that corresponds to the following impulse responses:

$$h(t) = \delta(t-1) + 2\delta(t-2)$$

$$h(n) = \begin{cases} \frac{1}{3} & x = 0,1,2 \\ 0 & \text{otherwise} \end{cases}$$

10 DTFT versus DFT

For the signal

$$x(n) = \begin{cases} 1 & n = 0 \\ 0 & else \end{cases}$$

Compute $X(\omega)$ and $X(k)$ (use $N = 4$ for the DFT).